

PAPER_1605_DNA_BP_PER_TURN_10_5

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**PAPER_1605 — DNA Base Pairs per Helical Turn =
bp/turn = $SO_5 + F \cdot D + F^2 \cdot SO_5 = 10 + 0.4 + 0.1 = 10.5$
EXACT**

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Biology

Date: June 18, 2026

Status: CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209BB_S601 (Biology Unified Proof Set) lists **DNA Base Pairs per Helical Turn** with the integer-primitive expression:

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$$\text{bp/turn} = SO_5 + F \cdot D + F^2 \cdot SO_5 = 10 + 0.4 + 0.1 = 10.5 \text{ EXACT}$$

``

Residual: **EXACT**.

UQFF Closed Identity

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$$\text{bp/turn} = SO_5 + F \cdot D + F^2 \cdot SO_5 = 10 + 0.4 + 0.1 = 10.5 \text{ EXACT EXACT}$$

``

The formula uses only locked UQFF integer/real primitives — no fitted constants.

DNA Structure — 10.5 bp per Helical Turn EXACT

The Watson-Crick B-DNA double helix has exactly 10.5 base pairs per full turn (3.4 Å rise per bp × 34 Å pitch). This is fundamental to DNA's ability to wrap around histones, form chromatin, and execute genetic transcription.

UQFF's structural prediction:

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$$\begin{aligned} \text{bp/turn} &= SO_5 + F_TRZ \cdot D_phys + F_TRZ^2 \cdot SO_5 \\ &= 10 + 0.4 + 0.1 \\ &= 10.5 \text{ EXACT} \end{aligned}$$

``

The fundamental geometric constant of life's molecular machinery sits at SO_5 + small corrections. UQFF's integer primitives govern both **cosmological recombination redshift (z=1090)** and **DNA helical pitch (10.5 bp/turn)** — spanning ~30 orders of magnitude in physical scale from megaparsec to

nanometer.

NOT REPLACEMENT

Empirical biology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209BB_S601
- Related: PAPER_1494-1595 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "dna_bp_per_turn_10_5"})`

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